Some partial solutions and/or hints are given below. Watch out for typographical errors and misprints. If you catch an error, please let me know.

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1. **[SELF]** Derive, using 'integration by parts', the result

$$\int_0^\infty du\,u e^{-bu} = \frac{1}{b^2}\ .$$

Hint/Solution →

$$\begin{aligned}\int du\,u e^{-bu} &= u\int du\,e^{-bu} \;-\; \int du\left(\int du\,e^{-bu}\right)\\ &= u\left(-\frac{1}{b}\right)e^{-bu} \;-\; \left(-\frac{1}{b}\right)^2 e^{-bu} \;=\; -\frac{1}{b}u e^{-bu} \;-\; \frac{1}{b^2}e^{-bu}\end{aligned}$$

$$\begin{aligned}\Longrightarrow\quad \int_0^\infty du\,u e^{-bu} = \frac{1}{b^2} &= -\frac{1}{b}\left[u e^{-bu}\right]_0^\infty \;-\; \frac{1}{b^2}\left[e^{-bu}\right]_0^\infty\\ &= -\frac{1}{b}(0-0) \;-\; \frac{1}{b^2}(0-1) \;=\; \frac{1}{b^2}\end{aligned}$$

This integral is used twice in this problem sheet.

2. Normalization. Assume the constants α_i to be real and positive.

(a) [**10 pts.**] Consider a single particle confined to the half-line $x \geq 0$. The inner product is given by $\langle\phi|\xi\rangle = \int_0^\infty \phi(x)\xi(x)dx$. Find the constant α_1 so that the wavefunction

$$\psi(x) \;=\; \alpha_1\sqrt{x}e^{-x}$$

is normalized.

Hint/Solution $\rightarrow$

$$\langle\psi|\psi\rangle = \int_0^\infty |\psi(x)|^2 dx = \alpha_1^2 \int_0^\infty xe^{-2x}dx = \alpha_1^2 \times \frac{1}{4}$$

We used the integral treated previously.
In order to have a normalized wavefunction, we need $\langle\psi|\psi\rangle = 1$; i.e.,

$$\alpha_1^2 \times \frac{1}{4} = 1 \qquad \Longrightarrow \qquad \alpha_1 = 2$$

(b) [**5 pts.**] Consider a spin-1/2 (two-state) system. Find the constant α_2 so that the following state is normalized:

$$|\psi\rangle \;=\; \alpha_2 \begin{pmatrix} 2i \\ -1+i \end{pmatrix} \quad .$$

Hint/Solution $\rightarrow$
The norm is

$$\begin{aligned}\langle\psi|\psi\rangle \;&=\; \alpha_2^2 \Big((-2i) \quad (-1-i)\Big) \times \begin{pmatrix} 2i \\ -1+i \end{pmatrix} \\ &=\; \alpha_2^2 \left(|-2i|^2 \;+\; |-1-i|^2\right) \;=\; \alpha_2^2\,(4+(1+1)) = 6\alpha_2^2\end{aligned}$$

Normalization requires

$$6\alpha_2^2 = 1 \qquad \Longrightarrow \qquad \alpha_2 = \frac{1}{\sqrt{6}}$$

(c) **[10 pts.]** Consider a system with an infinite-dimensional Hilbert space, for which any state can be expressed as a linear combination of the infinite set of states $|\phi_n\rangle$, with $n = 0, 1, 2, \ldots$, which satisfy $\langle\phi_m|\phi_n\rangle = \delta_{mn}$. Find the constant α_3 so that the following wavefunction is normalized:

$$|\psi\rangle \;=\; \alpha_3 \sum_{n=0}^{\infty} \frac{(-1)^n}{3^n}\, |\phi_n\rangle \;.$$

Hint/Solution →

$$\begin{aligned}
\langle\psi|\psi\rangle \;&=\; \left(\alpha_3^* \sum_{m=0}^{\infty} \frac{(-1)^m}{3^m} \langle\,\phi_m|\right)\left(\alpha_3 \sum_{n=0}^{\infty} \frac{(-1)^n}{3^n}\, |\phi_n\rangle\right) \\
&= \alpha_3^2 \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} \frac{(-1)^m(-1)^n}{3^m 3^n} \langle\phi_m|\phi_n\rangle \;=\; \alpha_3^2 \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} \frac{(-1)^m(-1)^n}{3^m 3^n}\delta_{mn} \\
&= \alpha_3^2 \sum_{m=0}^{\infty} \frac{(-1)^{2m}}{3^{2m}} \;=\; \alpha_3^2 \sum_{m=0}^{\infty} \left(\frac{1}{9}\right)^m \;=\; \alpha_3^2 \frac{1}{1-\frac{1}{9}} \;=\; \frac{9}{8}\alpha_3^2
\end{aligned}$$

Normalization requires

$$\alpha_3 \;=\; \sqrt{\frac{8}{9}} \;=\; \frac{2\sqrt{2}}{3}$$

(d) **[5 pts.]** For a system with a 4-dimensional Hilbert space, find the constant α_5 so that the following wavefunction is normalized:

$$|\psi\rangle \;=\; \alpha_4 \begin{pmatrix} -2 \\ -1+3i \\ i \\ -i \end{pmatrix} \quad .$$

Hint/Solution →

$$\begin{aligned}
\langle\psi|\psi\rangle \;&=\; \alpha_4^2 \left(|-2|^2 \;+\; |-1+3i|^2 \;+\; |i|^2 \;+\; |-i|^2\right) \\
&= \alpha_4^2 \left(4 + (1+9) + 1 + 1\right) \;=\; 16\alpha_4^2
\end{aligned}$$

Normalization requires

$$16\alpha_4^2 = 1 \qquad \Longrightarrow \qquad \alpha_4 = \frac{1}{4}$$

(e) **[SELF]** For a single particle confined the x-y plane, Find the constant α_5 so that the following wavefunction is normalized:

$$\psi(x,y) \;=\; \alpha_5 e^{-(x^2+y^2)/2} \; .$$

Hint/Solution $\rightarrow$

$$\begin{aligned}\langle\psi|\psi\rangle \;&=\; \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy |\psi(x,y)|^2 \\ &=\; |\alpha_5|^2 \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \left| e^{-(x^2+y^2)/2} \right|^2 \;=\; \alpha_5^2 \int_{-\infty}^{\infty} dx e^{-x^2} \int_{-\infty}^{\infty} dy e^{-y^2} \\ &=\; \alpha_5^2 \sqrt{\pi}\sqrt{\pi}\end{aligned}$$

Normalization requires

$$\alpha_5 \;=\; \frac{1}{\sqrt{\pi}}$$

3. Discrete energy values and blackbody radiation.

   (a) **[10 pts.]** In thermal equilibrium, the probability of a mode having energy E is proportional to $e^{-E/k_BT} = e^{-\beta E}$. Show that the average energy of the mode is $k_BT = 1/\beta$ if the energy E is a continuous variable, and is $hf/(e^{\beta hf} - 1)$ if the energy is discrete and quantized in units of hf.

   You might have to use the summation formulae

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \qquad \text{and} \qquad \sum_{n=0}^{\infty} n x^n = \frac{x}{(1-x)^2} \qquad \text{for } |x| < 1.$$

Hint/Solution →

If the energy E is a continuous variable, the average is

$$\frac{\displaystyle\int_0^{\infty} dE e^{-\beta E} E}{\displaystyle\int_0^{\infty} dE e^{-\beta E}} = \frac{1/\beta^2}{1/\beta} = \frac{1}{\beta}$$

If the energy is a discrete variable, $E_n = nhf$, the average is

$$\frac{\displaystyle\sum_{n=0}^{\infty} e^{-\beta E_n} E_n}{\displaystyle\sum_{n=0}^{\infty} e^{-\beta E_n}} = \frac{\displaystyle\sum_{n=0}^{\infty} e^{-\beta nhf} nhf}{\displaystyle\sum_{n=0}^{\infty} e^{-\beta nhf}} = \frac{hf \displaystyle\sum_{n=0}^{\infty} \left(e^{-\beta hf}\right)^n n}{\displaystyle\sum_{n=0}^{\infty} \left(e^{-\beta hf}\right)^n}$$

$$= hf \frac{e^{-\beta hf}}{(1 - e^{-\beta hf})^2} \Bigg/ \frac{1}{(1 - e^{-\beta hf})} = hf \frac{e^{-\beta hf}}{1 - e^{-\beta hf}} = \frac{hf}{e^{\beta hf} - 1}$$

(b) **[5+5 pts.]** Planck obtained the correct blackbody radiation spectrum by replacing the classical expression by the quantum expression:

$$\frac{1}{\beta} \longrightarrow \frac{hf}{e^{\beta hf}-1}$$

Show that the quantum expression reduces to the classical expression at small frequencies.

Hint/Solution $\rightarrow$
First method: taking the limit $f \rightarrow 0$ together with L'Hopital's rule:

$$\lim_{f\rightarrow 0}\frac{hf}{e^{\beta hf}-1} = \lim_{f\rightarrow 0}\frac{h}{\beta h e^{\beta hf}} = \frac{h}{\beta h\cdot 1} = \frac{1}{\beta}$$

Second method: At small frequencies, $e^{\beta hf} \approx 1+\beta hf$; so the quantum expression at small frequencies can be approximated by

$$\frac{hf}{e^{\beta hf}-1} \approx \frac{hf}{1+\beta hf-1} = \frac{1}{\beta}$$

(c) **[SELF]** Express the quantum expression above as a power series in f. The first term should be the classical expression. You could try.......

Hint/Solution $\rightarrow$
Let's work it out up to two further terms, using the binomial expansion:

$$\begin{aligned}
\frac{hf}{e^{\beta hf}-1} &\approx \frac{hf}{1+\beta hf+\frac{1}{2}\beta^2(hf)^2+\frac{1}{6}\beta^2(hf)^3-1} \\
&= \frac{hf}{\beta hf}\left(1+\frac{1}{2}\beta hf+\frac{1}{6}\beta^2(hf)^3\right)^{-1} = \frac{1}{\beta}\left(1+\frac{1}{2}\beta hf+\frac{1}{6}\beta^2(hf)^3\right)^{-1} \\
&\approx \frac{1}{\beta}\left(1-\left[\frac{1}{2}\beta hf+\frac{1}{6}\beta^2(hf)^3\right]+\left[\frac{1}{2}\beta hf+\frac{1}{6}\beta^2(hf)^3\right]^2\right) \\
&\approx \frac{1}{\beta}\left(1-\frac{1}{2}\beta hf+\frac{1}{12}\beta^2(hf)^2\right) = \frac{1}{\beta}-\frac{1}{2}hf+\frac{1}{12}\beta(hf)^2
\end{aligned}$$